Yang Sen (Liam) Lin | Curriculum Vitae/Resume

Program: Computer Science & Engineering M.S.E.

tommy60718.en10@nycu.edu.tw | github.com/tommy60718 | www.linkedin.com/in/yang-sen-lin

Education

National Yang Ming Chiao Tung University (NYCU) **Sep 2021 – Dec 2025 (expected)**
B.S., Mechanical Engineering & Computer Science (Cross-disciplinary) GPA: **4.08/4.3** (last 60), 3.85/4.3 (overall)
Academic Excellence Award × 2 (**Top 5% of Dept.**)
Ranking: **7/61 (11.4%)**

Relevant Coursework (AI, Vision & Algorithms): Data Structures & OOP (**A**), Intro to Algo (**A**), Computer Vision (Grad.) (**A+**), Intro to AI (**A+**), Database Systems (**A-**), CV for UAV Autopilot (**A**), JS Web Prog (**A**).

Relevant Coursework (Control, Math & Systems): Automatic Control I & II (**A+/A**), Engineering Math I & II (**A-**), Applied Electrics (**A+**), Smart Materials (**A+**), Applied Mechanics (Dynamics) (**A**).

Publications

[1] **See, Point, Fly: A Learning-Free VLM Framework for Universal Unmanned Aerial Navigation.**
CoRL 2025 (Accepted)
Chih Yao Hu*, **Yang-Sen Lin***, Yuna Lee, Chih-Hai Su, Jie-Ying Lee, et al. (* Equal Contribution)
*(Note: CoRL ranked **#3 in Robotics** on Google Scholar, **H5-index: 107**)*

Research Impact: Proposed a **zero-shot** navigation framework that decouples VLM reasoning from geometric control, achieving **SOTA** success rates in unstructured environments without task-specific training.
Technical Novelty: Formulated a **geometric projection algorithm** to ground 2D visual prompts into **3D metric controls**, solving the **depth ambiguity** problem inherently found in monocular vision setups.

Work & Research Experience

OMRON SINIC X *Tokyo, Japan*
Incoming Research Intern — Mentor: Dr. Masashi Hamaya *Jun. 2026 – Aug. 2026*
(Selected for a research position focusing on Robotics Foundation Models)

- **Proposed Research:** To formulate a knowledge distillation framework to compress large-scale Vision-Language-Action (VLA) models, aiming to resolve computational bottlenecks for **real-time inference** on **edge hardware**.
- **Research Goal:** To investigate **latency minimization** techniques for **high-frequency control loops** in soft robotics, with a target of submitting findings to top-tier conferences (e.g., **CoRL/ICRA**).

Computational Photography Lab, NYCU *Hsinchu, Taiwan*
Undergraduate Researcher — Advisor: Prof. Yu-Lun Liu *Sep. 2024 – Present*

- **Geometric Formulation (The Math):** Derived a **mathematical action-to-control mapping** to ground high-level VLM inference (2D coordinates) into low-level drone dynamics (3D trajectories), theoretically solving depth ambiguity without depth sensors.
- **Sim2Real Engineering (The Validation):** Overcame the simulation-to-reality gap by calibrating coordinate systems and **optimizing control latency**. Validated robustness through **55+ real-world flight experiments**, demonstrating long-horizon planning in unstructured environments.

Wolley Inc. *Hsinchu, Taiwan*
Firmware Engineer Intern — Advisor: Dr. Yu-Ming Chang (Director) *Jul. 2024 – Aug. 2024*

- **System Profiling (CXL/Memory):** Engineered C-based diagnostic tools to profile **throughput and latency** for **CXL Type-3** solutions, providing critical metrics to reduce **host bandwidth pressure**.
- **Parallel Computing & Optimization:** Refactored the internal testing system using **parallel computing** in Linux (optimizing I/O efficiency), which significantly reduced unit testing time for the main product line.

Technical Skills

- **Languages:** Python *(Proficient)*, **C/C++** *(System-level & Embedded)*, C# *(Unity)*.
- **AI & Computer Vision:** OpenCV, YOLO, **VLM Integration**, PyTorch, GANs.
- **Robotics & Control: PID Control**, UAV Dynamics, SDL *(Physics simulation)*.
- **Systems & Tools:** Linux *(Profiling)*, Git, Docker, **CXL/HDM Protocols**, Embedded Profiling Tools.

Selected Projects

Vision-Based UAV Autopilot System | *Python, OpenCV, Control Theory*

- **Algorithmic Optimization:** Optimized visual tracking algorithms from **quadratic** $\mathcal{O}(N^2)$ **to linear** $\mathcal{O}(N)$ **complexity** via **heuristic-based sparse sampling**, eliminating computational bottlenecks for real-time control.
- **Control Implementation:** Engineered a **closed-loop PID controller** from scratch, iteratively tuning proportional and derivative gains (K_p, K_d) to achieve **<3cm hover accuracy** without relying on external positioning systems.

Perceptual Image Restoration via High-Order Modeling | *Python, PyTorch, Signal Processing*

- **Mathematical Modeling:** Modeled signal degradation via **Generalized Gaussian kernels** $(D = (k \circledast I)_{\downarrow s} + n)$ to simulate complex blur artifacts.
- **Perceptual Alignment:** Mitigated domain shift by fine-tuning generative priors, achieving superior **Mean Opinion Scores (MOS)** in perceptual fidelity compared to standard interpolation baselines.

DesignMate: Computational AR Sketching Interface | *Unity, C#, Linear Algebra*

- **Geometry Pipeline:** Developed a **vector-to-mesh pipeline** using **cubic interpolation** to smooth discrete stylus inputs, driving a diffusion-based generation workflow.
- **Resource Efficiency:** Designed a robust polling mechanism to manage **GPU memory allocation**, preventing bottlenecks on **mobile AR hardware** during distributed inference.

Leadership & Service

Google Developer Student Club (GDSC) NYCU — *Hsinchu, Taiwan*
Chapter Lead — *Jul. 2024 – Jun. 2025*

- **Organizational Leadership:** Directed a 24-member team across 5 departments; supervised **6 AI & Software Engineering projects** bridging academic theory with industry practice.
- **Community Impact:** Orchestrated 10+ technical workshops, impacting **40+ student developers** and fostering a peer-learning ecosystem.

Meichu Hackathon (Logitech Division) — *Hsinchu, Taiwan*
Tech Lead & PM (3rd Place Winner) — *Sep. 2025*

- **Technical Management:** Led a cross-functional team of 5 to develop *DesignMate* under a **48-hour deadline**; **architected the system framework** based on team expertise.
- **Communication:** Delivered the final technical pitch, successfully communicating complex engineering concepts to non-technical judges.

NYCU Varsity Swim Team — *Hsinchu, Taiwan*
Team Member & Relay Gold Medalist — *Jan. 2022 – Dec. 2024*

- **Achievement of Grit:** Progressed from novice to **City-Level Gold Medalist** within 3 years. Balanced **8+ hours of weekly high-intensity training** with rigorous research commitments.

TEDxNYCU — *Hsinchu, Taiwan*
Selected Speaker — *Dec. 2023*

- **Community Leadership:** Selected to address an audience of 50+ on "Redefining Success"; utilized a structured narrative framework to articulate personal value systems, inspiring purpose-driven learning among university peers.

Community Service: Homeless Support Volunteer — *Taipei, Taiwan*
Initiator & Logistics Lead — *Sep. 2025 – Nov. 2025*

- **Social Initiative:** Initiated a 10-week meal distribution program; managed procurement logistics and led weekly reflection sessions to improve service delivery efficiency.

Honors & Awards

- **3rd Place**, Meichu Hackathon (Logitech Division) — *Sep. 2025*
- **2nd Place & Popularity Award**, Taipei Metro Hackathon *(Ranked 2nd out of 92 teams)* — *May 2024*
- **Gold Medal**, Men's 4x100m Relay, Hsinchu City Swimming Championship — *Sep. 2024*
- **Academic Excellence Award** $\times$ 2 *(Top 5% of Dept.)* — *Fall 2023, Fall 2024*